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INCORPORATING SYLLABLE STRUCTURE INTO THE TEACHING OF SPANISH PRONUNCIATION

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7.1 Introduction

This chapter presents a general overview of **syllable** structure in Spanish, focusing on those aspects that are relevant to the teaching of Spanish as an **L2**, such as **phonotactics**, **resyllabification** and **vowel merger**, **diphthongs** and **hiatuses**, the allophonic (see **allophone**) realizations of the high **vowels**, and **consonant** clusters. Overall, the approach to the phonological (see **phonology**) content is descriptive and atheoretical, although it relies on crucial phonological principles known to be widely present in both the **L1** and the **L2**, namely, syllable markedness (see **marked form**) and universal syllable repair mechanisms. In order to frame the suggestions offered for the integration of syllabic facts into the **L2** language classroom, the chapter includes a critical review of the treatment of the subject in recent, commonly used textbooks of Spanish pronunciation (Teschner 2000, Stokes 2005, Piñeros 2009, Schwegler, Kempff, and Ameal-Guerra 2010, Morgan 2010). It also highlights how knowledge of syllabic principles helps **L2** learners better their pronunciation, including references to previous studies on phonological awareness, and on the explicit teaching of this topic. Additionally, the chapter offers some suggestions on how to incorporate syllable structure into pronunciation training.

7.2 Syllable structure in Spanish: relevant phenomena for **L2** acquisition

7.2.1 The syllable and syllabic constituents

The syllable and syllabic structure are essential components of the phonological competence of a Spanish speaker and of the learning of **L2** Spanish phonology. Díaz-Campos (2014, 147), for instance, mentions 12 areas of phonological difficulty

for L2 learners of Spanish whose L1 is English. Of those 12, almost half (five) are related to **syllabification**. This chapter starts off with a descriptive overview of those syllabic phenomena that are relevant to the L2 learner of Spanish.

The syllable is a prosodic unit (see **prosodic constituent**) that consists of one or more **segments** surrounding a peak of **sonority** (i.e., segmental prominence), or **nucleus**. In Spanish the nucleus (N)—<a> in all three syllables in Figure 7.1—is always a vowel and is the only element required in a syllable. The consonantal elements preceding the nucleus are known as the **onset** (<p> and <g> in Figure 7.1), the non-vocalic segments that follow the nucleus are referred to as the **coda** (<r> in Figure 7.1) The **rhyme** (N') is a syllabic constituent that comprises the nucleus and the coda (<ar> in Figure 7.1).

In Spanish, any consonant may appear in the onset, but **palatal nasals** (see **nasal-stop class**) are rare (e.g., *ñandú*), and the **rhotic** tap [ɾ] does not occur as the sole member of the onset word-initially. The best coda consonants are two types of **sonorants**, nasals and **liquids**, and the **obstruent** /s/: *cambio*, *tren*, *Malta*, *papel*, *marca*, *amor*, *resto*, and *análisis*. Much less frequently, obstruents other than /s/ are also found in the coda, especially in word-final position: *concepción*, *obsesión*, *étnico*, *administrar*, *técnica*, *dogmático*, *carnet*, *reloj*, *tic-tac*, *pub*, and *chef* (Alonso 1945; Hualde 1989, 2005).

All syllabic constituents can be complex, meaning they can contain more than one segment. A maximum of two consonants is possible in the onset, while the rhyme allows up to three segments. An onset cluster must consist of an obstruent (i.e., /p/, /t/, /k/, /b/, /d/, /g/) or /f/ and a liquid /l/ or /r/ as the first and second members, respectively.¹ Coda segments exhibit more restrictions, well-formed combinations are: **glide** + consonant, *veinte* ['bejɲ.te] and *auxilio* [auk.'si]. *lio*, sonorant + /s/, *pers.pec.tiva*, *ins.tau.rar*, and *sols.ti.cio*, obstruent + /s/, *abs.te.ner* and *ads.cri.bir*. Coda clusters in which the second element is /s/ often delete the obstruent or the least sonorous consonant, especially in rapid and informal styles.

A well-formed nucleus consists of a vowel alone or a vowel and a glide in pre- or post-vocalic position (i.e., **rising** or **falling diphthong**, *miedo* ['mie].do, *peine* ['pei].ne). If the nucleus is preceded and followed by a glide, the resulting element is a **triphthong**, as in *buey* ['buej].

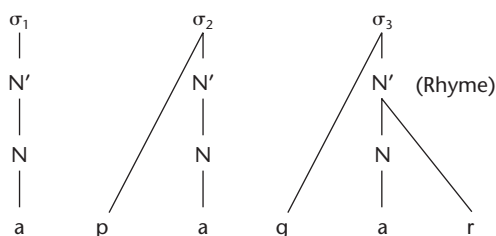


FIGURE 7.1 Structure of the Spanish syllable.

Source: adapted from Colina (2016, 26).

Example (1) presents examples of syllable types in Spanish (Colina 2009). The relevant syllable type is in **bold italics**.

(1) Spanish syllable types²

a. V	<i>a</i> .la	Ma.rí. <i>a</i>
b. CV	<i>la</i> .do	sa. <i>la</i> .da
c. CVC	<i>pan</i>	ca.pi. <i>tán</i>
CVG	<i>soy</i>	con. <i>voy</i>
CGV	<i>vió</i>	hun. <i>dió</i>
d. VC	<i>un</i>	<i>un</i> .tar
VG	<i>hay</i>	<i>Eu</i> .ro.pa
e. CCV	<i>flo</i> .tar	a. <i>fli</i> .gi.do
f. CCVC	<i>tren</i>	a. <i>pren</i> .der
CCVG	<i>plei</i> .te.ar	a. <i>plau</i> .dir
CCGV	<i>prue</i> .ba	em. <i>bria</i> .ga.do
g. VCC	<i>ins</i> .truir	re. <i>ins</i> .tau.rar
VGC	<i>aus</i> .tral	a.plau.di.rí. <i>áis</i>
h. CVCC	<i>pers</i> .pec.tiva	cir. <i>cuns</i> .tan.cias
CVGC	<i>cáus</i> .ti.co	to. <i>máis</i>
CGVC	<i>diez</i>	tras. <i>pies</i>
i. CCVCC	<i>trans</i> .por.te	re. <i>trans</i> .mi.tir
CCVGC	<i>claus</i> .tro	en.con. <i>tráis</i>
CCGVC	<i>truán</i>	san. <i>grien</i> .to
j. CGVC	<i>buey</i>	Pa.ra. <i>guay</i>

7.2.2 Syllabification

Native speakers have intuitions, quite strong for Spanish, about what constitutes a well-formed syllable and how to separate speech into syllables, a process known as *syllabification*. In Spanish, as in all languages, the most common syllable type consists of a consonant and a vowel (CV), therefore, an intervocalic consonant like *l* in *ala* is always parsed as an onset with the second vowel, as in *a.la*. This same principle applies across words, when a word ends in a consonant and the following starts with a vowel, they are parsed in the same syllable, with the consonant occupying the onset and the vowel being the nucleus. This is known as *resyllabification*; for instance, *las* and *amigas* are syllabified as *las* and *a.mi.gas* separately, but as *la.sa.mi.gas* in connected speech due to resyllabification. Additional examples are provided in (2).

(2) Resyllabification

a. <i>papel y pluma</i>	pa.pe. <i>ly</i> .plu.ma
b. <i>con atención y con esmero</i>	co. <i>na</i> .ten.ció. <i>ny</i> .co. <i>nes</i> .mero
c. <i>vamos a ver a los enfermos</i>	va.mo. <i>sa</i> .ve. <i>ra</i> .lo. <i>sen</i> .fer.mos
d. <i>vimos tres osos en el andén</i>	vi.mos.tre. <i>so</i> .so. <i>se</i> .ne. <i>lan</i> .dén

While there is a universal preference for CV, many languages do not resyllabify across words, which poses an issue for the L2 acquisition of Spanish.

Moving on to two consonants (CC), in word-medial position they can be syllabified together as a complex onset, as in *ha.bla*, or be divided into two syllables, one as the coda of the first syllable and the other as the onset of the second syllable, as in *can.sa.do*. Whenever possible (i.e., if the resulting cluster consists of an obstruent plus liquid /l/ or /r/), Spanish prefers a complex onset in word-internal position. This principle also applies to sequences of three consonants (CCC), where the options are C.CC, as in *com.ple.tar*, or CC.C, as in *ins.ta.lar*. In the case of four consonants (CCCC), the only possible parsing is CC.CC given that the maximum size of an onset cluster in Spanish is two (e.g., *ins.truir*).

Unlike intervocalic consonants, consonant clusters are not affected by resyllabification across words, even if the resulting cluster is a well-formed one, for example, *a.flo.jar* versus *chef.lo.co* **che.flo.co* (versus *chef.fla.co*). In other words, consonant clusters in the onset are not possible when the two consonants belong to different words.

The syllabification of vowel sequences is, as we will see in Section 7.2, a crucial element for the L2 learner. A series of two or three vowels can be pronounced in one syllable (i.e., diphthong or triphthong) or two syllables (i.e., hiatus). A diphthong consists of a glide (i.e., non-syllabic **vocoid**) in pre- or post-vocalic position, such as in *miedo*, *peine*, *suave*, and *autor*, while a hiatus has two full vowels (i.e., syllabic vocoids), such as in *ma.re.a* and *ha.bí.a*.³ In some Spanish varieties, the least sonorous vocoid is always realized as a glide, independent of its position in the syllable.⁴ In other words, sonority is the criterion for selecting the non-nuclear element of a diphthong: *m[íé].do*, *p[éí].ne*, *m[ué].ve*, and *[eu].nu.co*. One exception to this principle is that of stressed vocoids, when the least sonorous vocoid is stressed, the result is a hiatus, as in *Ma.rí.a* (but not *Ma.rio*). Dialectal variation must be considered with respect to vowel sequences and diphthongs. Not all Spanish dialects rely on sonority to select the non-syllabic element of a diphthong. Some US Spanish varieties, for instance, rely on syllabic position and select the second vocoid as the nucleus of the diphthong, as shown in (3) in the following ways: if the first vocoid is high, the resulting glide is high (3a), a mid vowel [e] or [o] is raised to high and pronounced as a glide (3b–c), when the first vowel is low, it is deleted (3d).⁵

(3) Glides versus vowels in diphthongs (Martínez-Gil 2000)

Peninsular Spanish ⁶	Spanish of the USA	
a. m[íə].mi.go	m[íə].mi.go	<i>mi amigo</i>
b. s[eu].sa	s[iu].sa	<i>se usa</i>
c. teng[oí].po	teng[ui].po	<i>tengo hipó</i>
d. l[ai].gle.sia	l[i].gle.sia	<i>la iglesia</i>

The process of diphthong formation (also known as *vowel merger*) also applies across words, when two vowels come in contact in connected speech (a vowel-final word followed by a vowel-initial one), as seen in the examples in (3) (e.g., *m[i.a].mi.go*

versus *m[ia].mi.go*). This process is subject to much stylistic variation, as the resulting configurations are affected by sonority, the stressed or unstressed nature of vowels, and pauses. In addition, all hiatuses can become diphthongs in connected speech through gliding (i.e., a process by which a vowel becomes a glide), for example, *ma.r[é.a]* versus *la.ma.[ɾia]* *sube*.

Some vowel sequences can exceptionally be pronounced as hiatuses, despite meeting the conditions for a diphthong, as in *r[i.é]n.do* versus *s[ié]ndo*. This has only come to the attention of phonologists over the last two decades. Many of these hiatuses are morphologically related to forms in which the high vowel carries stress, such as in *vi.a.ble* (*vi.a*), *ri.a.da* (*rí.o*), or involve a morphological boundary, as in *bi.enio*. A few, however, cannot be explained through morphological criteria and are usually considered exceptional: *dien.te* versus *cli.ente*, *miel* versus *ri.el*, and *fui.mos* versus *hu.i.mos* (Hualde 2005, 2014).

Syllabic structure also serves as a conditioning element for phonological phenomena in Spanish, such as: **nasal velarization** *pon* ['poŋ], **neutralization of place of articulation** in nasals and **laterals**, as in *alto* ['aɫ.ɬo] and *enfermo* [enj.'fer.mo] (see Chapter 6), voice and continuancy (see **continuant**) **assimilation** and neutralization in obstruents, as in *mismo* ['miz.mo], *pasta* ['pas.ɬa], and *obstáculo* [ops.'ɬa.ku.lo]/[obs.'ɬa.ku.lo]/[oβs.'ɬa.ku.lo] (see Chapters 4 and 13), **/s/-aspiration**, as in *estos* ['eh.ɬoh] (see Chapters 4 and 9), lateral and obstruent gliding, as in *volver* [boi.'βei] and *apto* ['aɫ.ɬo], and onset glide strengthening, as in *comiendo* [ko.'mieɲ.ɬo] versus *creyendo* [kre.'jeɲ.ɬo]. Knowledge of syllabic structure can be useful as an awareness-raising tool for students to understand the motivation shared by these processes, namely, the status of the coda as the most marked syllabic constituent. Given the marked status of codas, some languages do not allow them, while others allow some and restrict the featural specification of others, thus neutralizing (i.e., reducing) the number of possible phonological **contrasts**. Often the type of neutralization has to do with the nature of the segment in coda position; for instance, for nasals and laterals, neutralization affects the place of articulation in that these consonants can assimilate their place based on the place of the following onset consonant, as in *imposible* i[m.p]osible, *enfrentar* e[m.f]rentar, *tanto* ta[ɲ.ɬ]o, *Enrique* E[n.r]ique, *poncho* po[ɲ.ɬ]o, *cónyuge* co[n.j]uge, and *tengo* te[ŋ.g]o. In contrast, obstruents retain their place of articulation (with the exception of /s/), but show neutralization or assimilation of major class features such as continuancy and voice, thus surfacing as stops, fricatives, or approximants that are voiced or voiceless, as seen in (4). They are often deleted in casual and rapid speech.

- (4) a. /fuDbol/ ['fuɔ̯.βol] ~ ['fut.βol] ~ ['fu.βol] *['fup.βol] *['fuβ.βol]
 b. /oBsoleto/ [ob.so.'le.ɬo] ~ [oβ.so.'le.ɬo] ~ [op.so.'le.ɬo] ~ [oΦ.so.'le.ɬo] ~
 [o.so.'le.ɬo]
 *[oɔ̯.so.'le.ɬo] *[oɫ.so.'le.ɬo]

The onset is a preferred and strong syllabic constituent, therefore, it is not surprising that segments in this position are affected by strengthening processes. An

example of this in Spanish is the alternation in which a high glide, [i̯], is a consonant, [j], when parsed in the onset. We observe this when comparing (5a)/(5b) with (5c)/(5d).

- | | | |
|--------|---------|----------------|
| (5) a. | perd-er | per.d[i̯é].ron |
| b. | com-er | co.m[i̯é].ron |
| c. | cre-er | cre.[jé].ron |
| d. | o-ir | o.[jé].ron |

A final consideration in relation to syllabification that is crucial for L2 learners is that Spanish is a **syllable-timed** language, that is, syllables have approximately the same duration, regardless of their stress status (i.e., stressed or unstressed). This is in contrast with **stress-timed** languages, such as English, in which stressed syllables occur at regular intervals, regardless of the number of intervening syllables. One important consequence of this fact is that Spanish nuclear vowels always retain their quality and duration, and are not centralized into a **schwa** [ə] or [ɪ], nor deleted, regardless of whether they are stressed or not, for example, [a. 'me.ri.ka.] *[a. 'me.ri.kə] *['me.ri.kə] and [er. 'ma.no] *[er. 'ma.nə]. The only **vowel reduction** process to consider in Spanish is loss of syllabicity, which is seen in diphthongs (within words, e.g., *m[i̯é]do*) and syllable merger (across words, e.g., *m[i̯a].mi.go*). This phenomenon that does not entail vowel quality reduction or deletion, even for unstressed vowels (e.g., [*aj*]res *[*a*]res).

7.3 L2 acquisition of Spanish syllable structure

7.3.1 General issues

The purpose of this chapter is not to review general issues or models of L2 acquisition, but rather to focus on problematic aspects of Spanish syllable structure for the L2 learner, as identified by previous SLA research. Nonetheless, some basic notions of SLA theory need to be introduced in order to frame the discussion, to discriminate among the difficulties faced by the learner, and to refer to acquisition issues in general (versus language-specific ones).

L2 research clearly identifies two major factors involved in L2 acquisition, namely, L1 transfer and **developmental factors**, which involve language universals, markedness, and additional cognitive mechanisms. Most recent models of L2 acquisition recognize the role of both transfer and developmental factors (and others, such as age, motivation, etc.) in acquisition; however, less is known about how they interact. A significant body of research has been devoted to studying the role of transfer and developmental factors. Regarding syllable structure, Hancin-Bhatt and Bhatt (1996), for instance, discuss the interaction of transfer and developmental effects in the construction of English L2 syllable structures by Spanish and Japanese learners. Major studies English L2 clusters as produced by Brazilian learners to test the **Ontogeny Model** (Major 1994), which proposes that transfer

dominates acquisition during its beginning stages, while developmental factors slowly become more important as acquisition/learning progresses (Major 1986). Another example of research on transfer and universal factors in syllable structure is work by Carlisle (1988, 1998) that considers epenthesis and consonant clusters. Recently, formal models of phonology, such as **Optimality Theory**, have offered ways to capture the interaction of universal and language-specific transfer effects through the interaction of universal constraints ranked in language-specific hierarchies. In this model, transfer effects in the L2 reflect the L1 ranking of universal constraints, and developmental factors are the result of changes to the L1 ranking in order to approximate the ranking of the L2, in conjunction with universal constraints. Universal constraints (e.g., syllables are **open**) are said to be responsible for L2 forms that result from L1 inputs inexistent in the L2 (e.g., borrowed forms) but that also differ from L1 forms (Broselow, Chen, and Wang 1998).

Another area of research focuses on predicting the degree of difficulty encountered by learners. The **Differential Markedness Hypothesis** (Eckman 1977, 1981), for example, predicts that typologically marked structures and L2 elements absent in the L1 will be more difficult to acquire by learners. The **Speech Learning Model** (Flege 1995) argues that it is possible that a new category for an L2 sound differing from those of the L1 can be established and that the difficulty involved depends on the perceived difference between the target L2 sound and existing categories of the L1: the more distinct the sound, the easier it is to form a new category. Empirical studies also constitute a good source of information regarding areas of difficulty for specific L1/L2 combinations and can be used to test the predictions of theoretical models; for example, according to the Differential Markedness Hypothesis, it is expected that Spanish coda obstruents will be easier to acquire for English learners than for Portuguese learners, since English allows for more obstruent codas and coda clusters than Spanish and because Spanish also has more complex codas than Portuguese (Siqueira de Marrone 1990; Hualde 2005).

7.3.2 Problematic syllabic phenomena for L2 learners

In terms of problematic areas of the acquisition of syllable structure, overall, shorter syllables are preferred, as are single constituents over complex ones (e.g., singletons are preferred over complex onsets and monophthongs over diphthongs). Furthermore, longer structures tend to be reduced and the acquisition of more marked constituents implies the acquisition of shorter, less marked ones. Finally, constituents that meet the *sonority sequencing principle* are produced more accurately (Hansen 2009 and references therein).⁷ As mentioned above, Lord (2005) and Díaz-Campos (2014) highlight issues of pronunciation that are typically troublesome for Spanish learners and that are often included in pedagogical materials dedicated to the L2 pronunciation of Spanish. Among them, several are related to syllable structure, namely, the syllabification of vowel sequences within and across words, diphthongal vowel quality, consonant clusters, vowel reduction in unstressed syllables (i.e., syllable duration), and velarization of /l/ in syllable-final position (e.g., *papel* as

[pa.'peɫ], see **dark l** and Chapter 5 of this volume). This section reviews extant research on these issues, including mention of studies that refer to the effects of instruction. Because transfer and L1 influence play such prevalent role in L2 acquisition, it is difficult to discuss the acquisition of syllabic structure without making reference to specific language pairs. Since much of the research and teaching of Spanish pronunciation involves learners who have English as their L1, unavoidably, a significant part of this contribution will deal with English learners. Nonetheless, other languages will be mentioned when relevant research exists (e.g., Portuguese, Fang).

7.3.2.1 Syllabification of vowel sequences: Diphthongs and hiatuses

Acoustic (see **acoustic phonetics**) studies have shown that the pronunciation of Spanish diphthongs by L2 learners, in particular English speakers, is problematic and non-native-like (Parrondo Rodríguez 1999; Lord 2005; Zárate-Sáñez 2011; MacLeod 2012). One difficulty is that English does not have rising diphthongs because prevocalic glides (i.e., onglides) are parsed in the onset rather than in the nucleus. Furthermore, in American English, **homorganic** CCV clusters are not allowed, thus, the clusters /bw, pw, fw/ (labial) and /tj, dj, nj, lj, sj, zj/ (**coronal**) are not possible (Davis and Hammond 1995). As a consequence, Spanish rising diphthongs preceded by a homorganic consonant are predicted to become hiatuses in the L2 Spanish of American-English learners. Parrondo Rodríguez (1999) found confirmation for this prediction. In her study, all groups of learners performed quite well with [ɥe] diphthongs preceded by a labial consonant and in an open syllable (e.g., **pueblo**, **buena**), showing improvements in performance as proficiency increased. The same diphthong in a **closed syllable**, as in **Cuenca**, however, was more problematic, with a higher presence of hiatuses (i.e., CV.VC versus the correct CGVC). Spanish diphthongs in which the prevocalic glide was [i] yielded the worst performance by all subjects, that is, they showed a clear preference for hiatuses. In addition, in longer words, Parrondo Rodríguez (1999) observed deletion of the glide as well as parsing of the consonant in the previous coda, as in *ca.len.te* or *cal.ien.te* for *ca.lien.te*. Although she attributes this difference to cognitive overload, it could be stress related, as longer words in English are likely to have more vowel reduction. Finally, this author reported that in the case of diphthongs preceded by a consonantal cluster (i.e., CCGV, **Gabriel**, **griego**), the cluster did not have a significant influence on performance, and learners tended to favor a hiatus, as with single consonants; beginners, however, showed more of a tendency to attach the first consonant to the preceding coda, as in *Gab.riel*.

MacLeod (2012) also found empirical confirmation that English learners produce more hiatuses than native speakers of Spanish (e.g., *p[i.é].dra* for *p[íé].dra*), as indicated by longer vowel duration. Furthermore, her study showed that students transfer the phonological constraint against homorganic CG clusters, since glides in homorganic CG clusters were significantly longer than those in non-homorganic clusters in the speech of beginning learners, for example, *p[u.é]r.to* instead of *p[íé]r.to* (MacLeod 2012, 103). MacLeod's findings are further supported by Krause

(2013), who showed that L2 learners replace diphthongs with hiatuses and that learners have more success with diphthongization as their level of acquisition increases. Her subjects exhibited this behavior with real word data, but the non-sense word data did not show a similar relationship.

Similar results were obtained by Zárate-Sánchez (2011) who reported that Spanish learners tend to prefer a hiatus for vowel sequences that native speakers treat as a diphthong. Zárate-Sánchez's research focused on cross-linguistic influence (i.e., transfer) by studying the difference between cognates and non-cognates with regard to diphthong formation. He explains that the difference between Spanish learners and native speakers seems to be, at least in part, due to the inhibitory effect produced by cognates, particularly when vowel sequences corresponds to hiatuses in English (e.g., *mater*[iá]l in Spanish versus *mater*[i.ə]l in English), revealing a rather strong effect of transfer. His results indicate that diphthong formation is stronger when cognates are factored out. Spanish learners' intuitions for diphthong resolution were statistically the same among the three proficiency levels examined. Zárate-Sánchez (2011) speculates that this result has to do with the type of knowledge investigated, namely, implicit knowledge. The implicit knowledge for this particular target form is rudimentary because the presence or absence of a diphthong is not among the most salient features in learners' input. In the absence of the necessary knowledge, participants fall back on their L1, and probably on universal processes of acquisition as well, in order to resolve the structure of vocalic sequences. In view of these results, Zárate-Sánchez (2011) then examined the data from participants who were first excluded from the study because they reported having been taught Spanish syllabification rules and, therefore, were not using intuition but rather explicit knowledge to complete the task. The scores of these participants were significantly higher than those of participants who seemed to have used intuition only, both in cognates and non-cognates. These findings suggest that conscious knowledge of Spanish syllabification rules indeed aided these students in identifying the vocalic sequences as **tautosyllabic**, and therefore, that the pronunciation of vowel sequences in L2 learners of Spanish benefits from explicit instruction.

Many believe that phonological awareness and explicit teaching are crucial to the acquisition of L2 Spanish phonology (Arteaga 2000; Lord 2005; Morin 2007). This is supported by empirical findings. While some research studies only the pronunciation of consonantal segments (Castino 1996; Kissling 2015), there is no reason to believe that pronunciation improvements due to explicit instruction would be limited to the segmental domain. In fact, Lord found that her subjects' pronunciation of diphthongs improved as a result of instruction, especially across words (2005, 564). Learners in Lord (2008) also demonstrated more accurate pronunciation of vowels, diphthongs, and linking (i.e., diphthongs across the word). Pronunciation gains were reported by Elliott (1997) as well. In addition to improvement in the pronunciation of various segments, Elliott's findings showed that vowel centralization to schwa decreased as a result of instruction. I return to the topic of pronunciation instruction and teaching in Section 7.4, when I discuss techniques and suggestions for the teaching of Spanish syllabification.

Regarding the very scarce literature on languages other than English, L1 speakers of Fang (African language of Equatorial Guinea) resolve across-the-word vowel sequences through deletion of the first vowel (Bibang Oyee 1990, 2002). Vowel sequences are rare, if not impossible, in Fang. However, a more recent empirical study (Vuskovich 2013) found that in fact L1 speakers of Fang also used other strategies, such as V2 deletion and glide formation. Overall, however, subjects retained most vowel sequences. The author hypothesized that the immersion situation in which subjects learned the L2 may have resulted in improved production.

7.3.2.2 *Syllabification of consonant sequences*

Another area of syllabification that could pose problems for L2 learners of Spanish has to do with consonant clusters. Fang does not have C + rhotic onset clusters, only /Cj/ clusters. On the basis of sonority (namely, the least sonorous C is the best onset), it is predicted that learners will delete liquids; however, Vuskovich (2013, 190) found that the majority of participants in his study produced C + rhotic clusters in a manner that is very similar to the native speaker standard. This is probably due to the advanced proficiency of the learners. In cases where the cluster was not produced as it is in the L1, the results showed trilling of the flap. Vuskovich argues that this is a result of articulatory (see **articulatory phonetics**) timing issues, and therefore, has to be a developmental factor, as this phenomenon is not found in the L1 and has been reported in the L2 output of other learners.

Consonant clusters do not pose serious difficulties for English learners.⁸ This is not unexpected since these consonant groups are much more complex in English than in Spanish. Nonetheless, some issues remain as one considers the interaction of these clusters with other segments; for instance, as mentioned above, the homorganicity restriction on CG clusters in English (e.g., *cute* [kjut] but *tan* *[tjæn], MacLeod 2012) has an effect on the inaccurate production of diphthongs as hiatuses in L2 Spanish. Teachers often report problems with the parsing of consonant sequences word internally in that the Spanish sequence /sCC/, a possible onset in English but not in Spanish, is syllabified by L1 English learners in the onset, as in *[ˈra.sʔro] instead of [ˈras.ʔro].

For learners with Portuguese as the L1, Unger and Zappa (2009) reported inaccurate production of some coda consonants, evidencing transfer effects from the L1, such as coda liquids realized as glides, /fácil/ as [ˈfa.siu], as well as [i] epenthesis after a coda obstruent, [a.ʔi.ˈmoh.fe.ra], [a.ʔ.ˈmoh.fe.ra]. Pereira Machado and Bassols Brisolará (2010) reported gliding of coda liquids in the L2 Spanish of Brazilian learners, which tended to improve/decrease as a result of instruction.

Most Spanish pronunciation textbooks identify resyllabification as an area of difficulty for L2 learners, in particular, for L1 English speakers. A recent study showed that resyllabification also affects perception, especially word recognition and spoken language comprehension, as mapping phonetic forms to meanings is further complicated in L2 Spanish by resyllabification across word boundaries (e.g., *es uno* [ˈe.ˈsu.no], Estes 2014). Estes' results indicate that consonant-initial words were

identified at a higher rate than vowel-initial words. In addition, proficiency level and word class were significant factors, with content words identified at a higher rate than function words. Speech rate was not found to be significant in the study, despite the commonly held belief that difficulty with L2 listening is due to native speakers talking “too fast.” Estes argues that when speech rate is blamed, the problem is often the listener’s cognitive processing load, which is inversely related to proficiency. Segmentation, complicated by resyllabification and vowel merger, was identified as a factor related to perception by Morin (2007) and Parrondo Rodríguez (1999, 125).

7.3.2.3 *Unstressed vowel reduction*

Finally, unstressed vowel reduction is another problematic area in the pronunciation of L2 Spanish learners who have English as their L1, as indicated by coverage of this topic in most, if not all, pronunciation textbooks. Empirical studies (e.g., Menke and Face 2010) confirm this position (for further discussion, see Chapter 1 of this volume). Their findings reveal that bilingual Spanish-English speakers exhibited centralization tendencies when unstressed vowels were compared to their stressed counterparts. Of the five vowel **phonemes**, /a/ was the one that posed the greatest challenge with respect to reduction, possibly due to the fact that it is a central vowel in Spanish. In contrast, the native speaker control group did not demonstrate significant centralization or reduction of this particular phoneme in unstressed syllables.

7.4 Classroom teaching of Spanish syllabification

This section reviews textbook treatment of some of the issues identified as problematic in Sections 7.2 and 7.3 and includes some suggestions as to how to address them in the classroom through sample lessons designed by the author.

Most recently published English-learner textbooks, if not all, include syllabification among their teaching units. Some of the most popular ones (i.e., Teschner 2000; Stokes 2005; Piñeros 2009; Morgan 2010; Schwegler, Kempff, and Ameal-Guerra 2010) cover the major topics identified in Section 7.2: syllabification of consonants and consonant groups (onset and coda, onset and coda clusters), syllabification of vowels and vowel sequences, both within and across the word (i.e., diphthongs, hiatuses, and vowel mergers), and full vowels versus vowel reduction/schwa. This is not surprising since the identification of problematic areas of pronunciation has traditionally relied on teachers’ experience rather than on empirical research, which only started to experience significant growth over the last two decades, mainly as a result of technological advances in speech recording and analysis software. Empirical findings generally corroborate the conclusions of descriptive studies and teachers’ experience with syllabification problems encountered by L2 learners (cf. Section 7.3); however, they also highlight specific elements of syllabification that explain the source of learners’ difficulties, such as the restriction on

homorganic onset clusters in English as the source of hiatuses in L2 Spanish (cf. Section 7.3.2.1), which are rarely covered in textbooks (if at all).

Teaching manuals differ slightly from one another with regard to the extent and technical nature of the presentation of syllabification topics in terms of their target readership and the teaching activities they include. The textbooks referred to above range from Stokes (2005), a shorter, less technical book (aimed mostly at students in their third year of Spanish) on one end of the continuum, to Piñeros (2009) and Schwegler, Kempff, and Ameal-Guerra (2010) on the other end, both of which are more extensive and technical in their treatment of syllabic phenomena (across-the-word vowel combinations, in particular); for instance, Piñeros explains the notion of syllabic sonority, a concept that is not addressed in any significant manner by the other books mentioned and that is likely to be better received by students specifically interested in the phonology and **phonetics** of Spanish.

These textbooks use many of the awareness-raising suggestions that Lord (2005) found to be beneficial overall, such as: explanations of different syllabification principles of English and Spanish segments, oral and listening practice, and transcription practice. Awareness-raising activities have traditionally played a privileged role in Spanish textbooks, given that, as Stokes puts it, “we often focus so much on our message as we speak, that it is often challenging to pay close attention to how we sound (2005, xii).” Some of the above-mentioned textbooks are also innovative in their frequent use of poems and aphorisms, tongue-twisters, and contextualized, communicative pronunciation practice to teach syllabification, especially across-the-word phenomena (Stokes 2005; Morgan 2010). Poetry is particularly relevant, as it merges across-the-word vowel sequences to obtain the required number of syllables for a line (i.e., **synalepha**). Although most books include poetry at some point, Stokes makes extensive use of poems to practice the syllabification of vowel sequences. He also encourages students to record and analyze their own speech, as well as samples of others. Other useful classroom techniques highlight contrasts between English and Spanish syllabic phenomena by having students decide whether a speaker is using an English schwa or a full Spanish vowel (/o/, /a/) (e.g., Schwegler, Kempff, and Ameal-Guerra 2010). Popular textbooks also reflect Estes’ (2014) assessment that perception has played a much less prominent role in the classroom than production. While it is true that listening to recordings is a significant element of practice, listening is more often resorted to as part of a listen-and-repeat or transcription activity rather than a purely perceptual exercise where students’ attention is focused on perceiving differences between target pronunciations and inaccurate ones (e.g., resyllabification versus hiatus or **glottal stop** (see **stop consonant**) insertion, hiatus versus diphthong, schwa versus full vowel).

Finally, I conclude with a few suggestions of my own and a couple of sample lessons. I am not aware of any empirical testing of these suggestions, but they have worked well for me and my students and are supported by some of the empirical findings reported above with regard to the acquisition of L2 Spanish phonology.

Estes (2014) mentions the use of videos as a technique to raise awareness and to focus on perception (in addition to production). Music videos can be used for this

purpose, and to focus on relevant aspects of Spanish pronunciation presented in class, in this case, syllable structure. Moreover, music videos are excellent for students to sing along to and to perceive natural language without relying so heavily on orthography and written text.⁹ In addition, they include multiple examples of connected speech and across-the-word phenomena.

My first sample lesson focuses on syllabification using music videos. Its objective is to improve the perception and production of the syllabification of vowels across words, as well as consonant resyllabification. I should note that this sample lesson uses a public-domain children's song for illustrative purposes, but many others are possible, in particular, using online free streaming sources. The first step of the lesson involves listening to a music video, such as the song *La vaca lechera* (see www.youtube.com/watch?v=KU9ju5s3ueY). Teachers should make sure that students listen to the video in class, and should select one that does not have the song's lyrics on the screen or that has the close-captioning off. Next, teachers identify portions of the song and their class sings them together. In this step, it is important that teachers identify fragments that contain vowel merger or resyllabification ahead of time, and have the students sing and repeat those sections. Students should not see said sections written out. Teachers should also tell students not to worry if they do not understand all the words, rather, they should just try to imitate what they hear as best as possible. Then, teachers can play those segments of interest again and students can transcribe what they hear and separate the syllables. Finally, students should write out the words in the highlighted segments and comment on their pronunciation in relation to what they have learned in class. Teachers can put the lyrics on the board/screen after giving students some time to write the lyrics. As a follow up, teachers can assign homework in which students find a video of their choice and locate examples of the same phenomena discussed in class. Students should come to the following class ready to present about the video they chose and the examples they observed. A sample text is presented in (6).

- (6) Possible text (with relevant segments bolded and their corresponding transcriptions below them)

Tengo una vaca lechera

[goɰ].na

No es una vaca cualquiera

['noɛ.su]/['nɛɛ.su] ['kiɛ.ra]

Me da leche condensada

Para toda la semana

Tolón tolón tolón tolón

Tolón tolón tolón tolón

Tengo una vaca lechera

[goɰ].na

No es una vaca cualquiera

['noɛ.su]/['nɛɛ.su] ['kiɛ.ra]

Se pasea por el prado

[ˈpa.sɛa] [po.rel]

Mata moscas con el rabo

[ko.nel]

Tolón tolón tolón tolón

Tolón tolón tolón tolón

Tengo una vaca lechera

No es una vaca cualquiera

Me da leche merengada

Ay que vaca tan salada

Tolón tolón tolón tolón

Tolón tolón tolón tolón

Tengo una vaca lechera

No es una vaca cualquiera

Hace torta de cereza

Ay que vaca tan traviesa

Tolón tolón tolón tolón

Tolón tolón tolón tolón

Variation of text

Tengo una vaca lechera

No es una vaca cualquiera

un cencerro le he comprado

[le:.kom]

y a mi vaca le ha gustado

[ja.mi] [lɛa]

Tolón tolón tolón tolón

Tolón tolón tolón tolón

Cognates and borrowings are also helpful to compare and contrast the same sequence of sounds for native speakers of the L1 and the L2, as Zárate-Sández (2011) suggests with regard to diphthongs; for instance, L1 English learners can be asked to pronounce/listen to the pronunciation of *barrio* in English versus Spanish. In terms of syllabification, students would ideally observe a hiatus in the first case and a diphthong in the second case. They can also be asked to discriminate between the English and the Spanish pronunciation of cognates, possibly with authentic or synthesized speech in which only the relevant features have been altered. My second sample lesson plan involves the use of cognates to teach Spanish syllabification to L1 English learners. Its primary objective is to help students' perception and avoidance of vowel reduction and hiatuses in cognates. First, teachers should record an early bilingual pronouncing the words in (7) in English and Spanish (the same speaker). Other cognates with a VV sequence can be added to the list in (7). Using

a digital audio application, teachers can segment the syllables with the vowel sequences in English and Spanish. Then they should play the segments to students in a random order and ask them to identify each example as English or Spanish, and to justify their selections. Teachers can later point out all examples of hiatuses and schwas. A couple of variations of this activity that would be effective to implement are as follows: 1. Teachers take the words in (7), record native English speakers pronouncing them in English and native Spanish speakers pronouncing them in Spanish, and ask students to identify the language of the speaker and the differences in syllabification, and 2. Teachers invite two simultaneous bilinguals (or two native speakers, one of English, one of Spanish) to class, place them behind a screen so that students cannot identify them, ask them to pronounce the words in (7) in English and Spanish, and ask students to identify the differences in syllabification.

(7) Word list

barrio
pueblo
patio
California
Indiana
adiós
ciénaga
fiesta
sierra

7.5 Conclusion

This chapter has presented a general overview of the teaching of syllable structure in Spanish, focusing on those aspects that research has identified as problematic, such as diphthongs and hiatuses, the allophonic realizations of high vowels, syllabification of consonant sequences, and unstressed vowel reduction. Although most current research focuses on the teaching of Spanish as an L2 to English learners, references to other languages were included as well. The chapter highlighted how knowledge of syllabic principles helps L2 learners better their pronunciation, including references to previous studies on phonological awareness and the explicit teaching of this topic. Additionally, the chapter offered some suggestions on how to incorporate syllable structure with pronunciation training.

Notes

- 1 The only exceptions are /dl/—ill-formed in all dialects—and /tl/, in some.
- 2 C = consonant, V = vowel, G = glide.
- 3 The term vocoid is used to refer to non-consonantal segments, both syllabic and non-syllabic (i.e., vowels and glides).
- 4 The most sonorous vocoid is /a/, followed by /e/, /o/ and /i/, and finally, /u/.

- 5 Dialectal variation is discussed here and in other parts of this chapter. While the issue may not at first seem relevant for a chapter on L2 pronunciation, pronunciation teachers in the US encounter it in their L2 classrooms due to growing numbers of **heritage speakers**, who speak a number of Spanish varieties, as a result, teachers need to be aware of dialectal issues in order to better address needs of students. Furthermore, true L2 learners are becoming increasingly exposed to non-classroom varieties of Spanish through service learning, internships, etc., and may be acquiring or becoming familiar with some of these features.
- 6 In formal, careful styles, *s[eɪ].sa* and *ten.g[oi].po* tend to be pronounced in hiatus, as *s[e.ɪ].sa* and *ten.g[o.i].po*, respectively, because the high vowel is stressed.
- 7 Syllables exhibit a contour based on the universal sonority scale, according to which syllables rise in sonority toward the nucleus (the most sonorous point) and then decrease toward the coda (Clements 1990). Segments are ranked according to sonority in the universal sonority scale: Obstruents < Nasals < Liquids < Glides < Vowels.
- 8 I refer here to the possible difficulties of parsing consonants in a syllabic component rather than to the difficulties that may affect the phonetic realization of the individual segments that make up the cluster.
- 9 As an anonymous reviewer points out, students often like to use lyrics to help them better understand sung speech and to sing along themselves.

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